

R - sqldf() returning dataframe with zero rows

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I am trying an efficient means of reading in select data from very large csv files using the method described [here](#) by @JD_Long. The method relies on `sqldf()` a function from a library by the same name. The problem I am having is that the function returns an empty dataframe, that is a dataframe with the correct number of columns and their names but with zero rows.

I've truncated the formatting function that's intended to process the dataframe once it's been read into the working environment by `sqldf()`.

```
sum_stats_df <- function(f_name){
  # read in data
  f <- file(f_name, open = "r")
  df <- sqldf("select * from f limit 6", dbname = tempfile(),
             file.format = list(header = T, row.names = F))
  close(f, type='r')
  return(df)
```

The csv datasets that I'm working with are the well-known Bureau of Transportation Statistics flights data which I downloaded from the ASA website [here](#).

I tried the sample code from the StackOverflow question linked above.

```
bigdf <- data.frame(dim=sample(letters, replace=T, 4e7), fact1=rnorm(4e7),
  fact2=rnorm(4e7, 20, 50))
write.csv(bigdf, 'bigdf.csv', quote = F)
library(sqldf)
f <- file("bigdf.csv")
bigdf <- sqldf("select * from f", dbname = tempfile(), file.format = list(header = T,
  row.names = F))
```

It worked just fine when I called `sum_stats_df("bigdf.csv")`. I tried looking at the properties for the simulation file and the flights csv files using properties in the windows file explorer but they look identical informat. The only difference is the file size. The flights files are much larger, but I don't see how this would matter.

The next thing I tried is loading twenty lines of one of the flights csv files using `read.csv(..., nrow=20)`, and then took the resulting dataframe and wrote it back to a csv file. When I tried the problematic function on that new csv test file, it worked.

```
> print(head(sum_stats_df("test.csv")))
  X Year Month DayOfMonth DayOfWeek DepTime CRSDepTime ArrTime CRSArrTime
UniqueCarrier
1 "1" 1987  10         14         3    741         730    912         849
"PS"
2 "2" 1987  10         15         4    729         730    903         849
"PS"
3 "3" 1987  10         17         6    741         730    918         849
"PS"
4 "4" 1987  10         18         7    729         730    847         849
"PS"
5 "5" 1987  10         19         1    749         730    922         849
"PS"
6 "6" 1987  10         21         3    728         730    848         849
"PS"
```

But, again, a call using the original csv file returns an empty dataframe.

```
> print(head(sum_stats_df("2000.csv")))
[1] Year      Month      DayOfMonth  DayOfWeek  DepTime
[6] CRSDepTime ArrTime    CRSArrTime UniqueCarrier FlightNum
```

```
[11] TailNum      ActualElapsedTime CRSElapsedTime  AirTime      ArrDelay
[16] DepDelay      Origin          Dest           Distance     TaxiIn
[21] TaxiOut       Cancelled      CancellationCode Diverted
CarrierDelay
[26] WeatherDelay  NASDelay       SecurityDelay   LateAircraftDelay
<0 rows> (or 0-length row.names)
```

So I'm wondering what could it be about the csv files that I downloaded from ASA that is different from those that I write locally using `write.csv()` ? What might be other causes for `sqldf()` returning empty dataframes, but getting all the column names?

r csv sqldf

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asked Jan 9, 2018 at 5:01



Conner M.

1,934 3 19 29

One problem I see is that you are using `LIMIT` without `ORDER BY`. This doesn't make sense because you aren't telling the database *how* you want to limit the result set. Which 6 records do you want? – Tim Biegeleisen Jan 9, 2018 at 5:06

As far as I know, and I might be wrong, using `limit` this way would be analogous getting just the first six rows. – Conner M. Jan 9, 2018 at 5:11

But I've tried it without `limit` and get the same empty dataframe. – Conner M. Jan 9, 2018 at 5:11

- 1 Please read [this accepted answer](#) and see if you can read from the `file` as you have opened it in your function. – Tim Biegeleisen Jan 9, 2018 at 5:17

Added `l <- readLines(con=f, n=6); print(l)` and got the header and five full lines from the file. I've read many answers here saying that the `sql` objects are deleted immediately after the call, but it doesn't explain why the results are first assigned to `df` nor why it works with locally written `csv` files. – Conner M. Jan 9, 2018 at 5:28

2 Answers

Sorted by: Highest score (default)



SQLite is fussy about end of line characters and these can differ from the default on your OS if the file is from somewhere else.

1

On Windows and Linux Ubuntu this worked for me assuming that you have already unzipped it:

```
read.csv.sql("1987.csv", "select * from file limit 3", eol = "\n")
```

or on Windows this variation using the original bz2 file and assuming that the 7z executable is on your path:

```
read.csv.sql("1987.csv", "select * from file limit 3", eol = "\n",  
  filter = "7z x -so 1987.csv.bz2 2>NUL")
```

On Linux we can use bzip2 instead:

```
read.csv.sql("1987.csv", "select * from file limit 3", eol = "\n",  
  filter = "bzip2 -cd 1987.csv.bz2")
```

FAQ 16 on the [sqldf github page](#) has an example that is very similar to this one.

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edited Jan 11, 2018 at 13:48

answered Jan 9, 2018 at 22:03



G. Grothendieck

251k 17 198 333

The files you can download from the quoted web page are bzip compressed. I can reproduce your problems when using the compressed file directly. One possible solution is to decompress the file(s) before using them with `sqldf`. This is unfortunate since other tools for reading CSV files support bzip compressed files directly. I do not know how to do this with `sqldf()` directly, but `read.csv.sql` from the same package supports a `filter` option, making the following function possible:

```
sum_stats_df <- function(f_name){  
  read.csv.sql(f_name, sql = "select * from file limit 6", filter = "bunzip2")  
}
```

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answered Jan 9, 2018 at 15:23



Ralf Stubner

26.1k 3 39 75

Thanks for your input, but I unzipped the files before trying to load them. I wonder, though, if there is some processing or formatting going on with the unzipper that might be behind the problem. – [Conner M.](#) Jan 9, 2018 at 16:11

Which tool on which OS did you use to unzi^{pp} the files? What file sizes do you currently have? For comparison, the data from 2008 is 109 MB with and 658 MB without compression. – [Ralf Stubner](#) Jan 9, 2018 at 16:20

Opening the locally created csv files and the downloaded and unzipped files in Excel doesn't reveal any difference in the two. But opening them in notebook++ showed some quotes for values in some columns in the former. With this in mind I tried defining the class for each column using the method arg. Still didn't work. – [Conner M.](#) Jan 9, 2018 at 16:42

I used 7_zip, and the unzipped 2008 csv file is 673MB in my case. – [Conner M.](#) Jan 9, 2018 at 16:44

This is very odd. Can you add `print(f)` to your function after opening the file and post the result when calling the function on one of the files? – [Ralf Stubner](#) Jan 9, 2018 at 20:52
